

Objective and Comparative Characterization of Magnetic Tape Emulations: Calibration Protocols, Response Maintenance vs. Gain, and Nonlinear Frequency Compression

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Abstract—An empirical and objectively reproducible measurement protocol is established to evaluate and compare free magnetic tape emulation plugins. The *Iron Deck* plugin (M Media Audio) is analyzed against a representative reference processor from an internationally established analog recording hardware firm. The research examines the divergence of equalization standards between both devices—coinciding only in NAB, while *Iron Deck* offers IEC/AES and the reference exposes CCIR—and evaluates physical calibration protocols (MRL/STL calibration tapes, magnetic fluxes of 200, 250, 355, and 510 nWb/m) and their correspondence in the digital domain. Through Python scripts, three exclusive analyses are performed: (1) spectral calibration stability under gain excursion (*drive sweep*); (2) static transfer curve Input→Output at multiple frequencies; and (3) frequency-dependent nonlinear compression. The results show that *Iron Deck* matches the digital integrity of the reference device in frequency stability and anti-aliasing, and offers versatility advantages in the selection of equalization standard and transport speed. This study sets a methodological precedent for the audio engineering community.

Index Terms—magnetic tape emulation, MRL calibration tape, NAB/IEC/CCIR equalization, frequency response, total harmonic distortion (THD), frequency-dependent nonlinear compression, aliasing, audio DSP, objective analysis.

I. INTRODUCTION

In the field of audio signal processing, analog emulations play a critical role by imparting dynamic cohesion and harmonic coloration characteristic of the magnetic medium. Historically, the appreciation of these processors has been subject to the commercial prestige of the developing brand or the name of the modeled physical machine, relegating objective analysis to the background.

This research addresses a rigorous comparison between two free magnetic tape emulations from opposite origins: *Iron Deck* (M Media Audio), a plugin from an independent developer with parametric control of speed and equalization curve; and a reference plugin distributed by a leading firm in analog recording hardware, treated anonymously here for reasons of technical independence.

The central objective is not to subjectively determine which device produces a more appreciated sound coloration, but to

build an empirical precedent based on reproducible DSP metrics: frequency and phase response, total harmonic distortion (THD), frequency-dependent nonlinear compression, spectral calibration drift, and aliasing immunity. All measurements were performed programmatically to guarantee their reproducibility independent of perceptual criteria.

II. PHYSICAL FOUNDATIONS AND CALIBRATION STANDARDS

A. Magnetic Losses and Head Geometry

The physics of tape transport imposes restrictions dependent on speed v and magnetic wavelength $\lambda = v/f$. At low speeds (7.5 ips), the wavelength of the highs (10 kHz) is drastically reduced, increasing gap losses:

$$L_g(f) = 20 \log_{10} \left| \frac{\sin(\pi g/\lambda)}{\pi g/\lambda} \right| \quad (\text{dB}) \quad (1)$$

where g is the physical width of the playback head gap. The Wallace separation loss [5], $L_d(f) = 54.6 (d/\lambda)$ dB, attenuates the highs when the tape separates from the head. In the low range, the geometry of the magnetic poles introduces constructive/destructive interference known as *head bump*, characteristic in the 50–100 Hz range.

B. Equalization Standards: Convergence in NAB and Divergence in IEC/CCIR

A fundamental finding of this study is the divergence of equalization standards between the two evaluated devices:

- **NAB** (National Association of Broadcasters): the only standard present in *both* plugins. It uses time constants $\tau_1 = 50 \mu\text{s}$ ($f_1 \approx 3183 \text{ Hz}$) and $\tau_2 = 3180 \mu\text{s}$ ($f_2 \approx 50 \text{ Hz}$), with a high-frequency boost and low-frequency emphasis (American standard).
- **IEC 1/CCIR** (European standard, 15 ips): available only in the reference device. Constant $\tau_1 = 35 \mu\text{s}$ ($f_1 \approx 4547 \text{ Hz}$), without low-frequency emphasis, producing a flatter and more transparent profile in the mid-high band.

- **IEC 2/AES (30 ips):** available only in Iron Deck. Constant $\tau_1 = 17.5 \mu\text{s}$ ($f_1 \approx 9095 \text{ Hz}$), designed for maximum extended response in high frequencies and lower noise floor at high speed.

The inclusion of CCIR in the reference reflects the modeling of European studio equipment (e.g., Studer, Telefunken), while the IEC/AES offering in Iron Deck targets high-resolution workflows at 30 ips. This design divergence is in itself a subject of technical study.

C. Physical Calibration Protocols and their Digital Mapping

In physical tape machines, studio alignment requires certified calibration tapes (MRL/STL) [4] to adjust:

- 1) **Azimuth:** perpendicularity of the playback head using a 10–15 kHz tone.
- 2) **Operating flux:** calibration of $0 \text{ VU} \equiv +4 \text{ dBu} \equiv -18 \text{ dBFS}$ with a 1 kHz tone at fluxes of 200, 250, 355, or 510 nWb/m.
- 3) **Bias adjustment:** polarization current of 100–200 kHz adjusted by *overbias* (−1.5 dB at 10 kHz) to optimize the balance between distortion and noise.

In the digital domain, this protocol is evaluated by verifying that the processing does not introduce frequency drifts or erratic phase shifts under gain excursions, which is the subject of Section IV-C of this work.

III. PROGRAMMATIC DSP ANALYSIS METHODOLOGY

Measurements were performed programmatically in Python 3 (`scipy.signal`, `numpy`) at 48 kHz / 32-bit float, on signals generated and processed by each plugin in real-time under the exact conditions of each experiment:

- **Frequency response:** logarithmic sine sweep (1 Hz–24 kHz). The transfer function $H(f)$ was estimated using the cross-spectral density (CSD) ratio:

$$H(f) = \frac{S_{xy}(f)}{S_{xx}(f)} \quad (2)$$

with Savitzky-Golay smoothing ($w = 151$, $p = 2$) to remove spectral noise without distorting the EQ slopes.

- **Static transfer and THD:** pure tones at 100 Hz, 1 kHz, and 5 kHz, evaluated at input levels of −30, −18, −12, −6, and 0 dBFS to characterize compression and total harmonic distortion (H_2-H_{19}).
- **Calibration stability:** reference tones at 3000 Hz (NAB pattern tone) and 3150 Hz (DIN/IEC), measured with Welch ($N_{\text{FFT}} = 4f_s$) for sub-Hz resolution.
- **Spectral distribution:** pink noise processed and segmented into 10 octave bands centered from 31.5–16000 Hz.

Input files follow the naming convention:

$$\{\text{Device}\}-\{\text{IPS}\}_{\text{IPS-EQ}_{\{\text{curve}\}}}-\{\text{P1}\}_{\text{V1}}-\{\text{P2}\}_{\text{V2}}. \Delta H(f, D) = |H(f, D)|_{\text{dB}} - |H(f, D_{\text{nom}})|_{\text{dB}} \quad (3)$$

which allows automatic extraction of metadata (speed, EQ standard, drive level) via regular expressions, without assuming a fixed parameter grid.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Nominal Tonal Portrait: Iron Deck

Fig. 1 presents the direct comparison of the spectral response $H(f)$ at nominal level (−12 dBFS) of Iron Deck between NAB (15 ips) and IEC (30 ips) modes. The NAB curve exhibits a *head bump* of approximately +0.8 dB at 75 Hz, characteristic of the NAB bass boost, with a progressive roll-off above 10 kHz. The IEC curve at 30 ips extends the linear response up to 22 kHz, with a considerably flatter behavior in the mid-high band. The differential panel $\Delta H(f)$ quantitatively isolates the gain or loss per band when switching between standards.

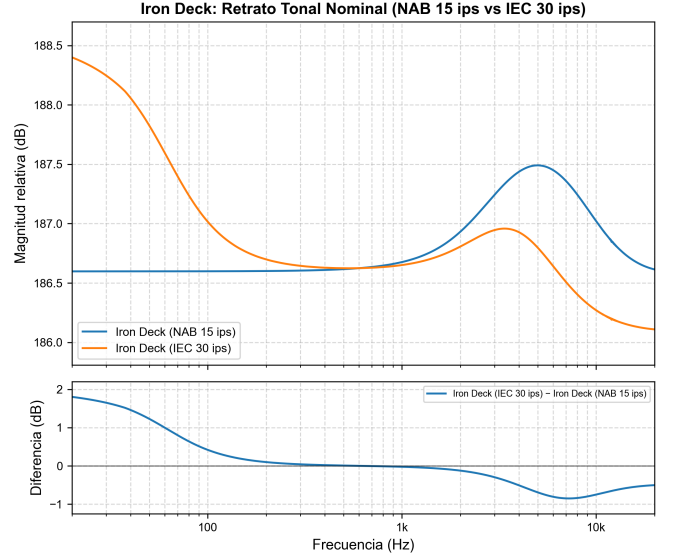


Figure 1. Nominal tonal portrait of Iron Deck. Comparison between NAB curve (15 ips) and IEC (30 ips) at −12 dBFS input level. The lower panel shows the spectral difference $\Delta H(f)$ (IEC−NAB) in dB.

B. Nominal Tonal Portrait: Reference Device

Fig. 2 presents the tonal portrait of the reference device comparing NAB (15 ips) and CCIR (15 ips). The CCIR curve shows progressive attenuation above 12 kHz and the absence of a bass boost, characteristic features of the European standard. The difference $\Delta H(f)$ relative to NAB exceeds −3 dB in the high-frequency bands, quantifying the tonal impact of the playback standard choice.

Figs. 3 and 4 detail the effect of transport speed (7.5, 15, 30 ips) and equalization curve (NAB vs. IEC) in Iron Deck, respectively.

C. Exclusive Analysis: Calibration Maintenance vs. Gain

To determine whether the saturation (*drive*) control alters the plugin’s frequency calibration, the relative tonal drift is defined:

where D_{nom} is the nominal operating point (REC control = 5 for Iron Deck, Input = 0 dB for the reference). Fig. 5 presents this drift for both devices at 15 ips, NAB.

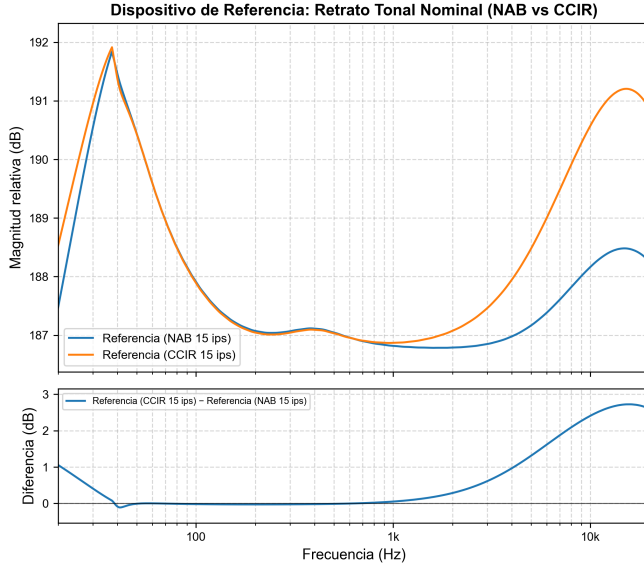


Figure 2. Nominal tonal portrait of the reference device. Comparison between NAB and CCIR (both at 15 ips, -12 dBFS). The lower panel shows $\Delta H(f)$ (CCIR–NAB) in dB.

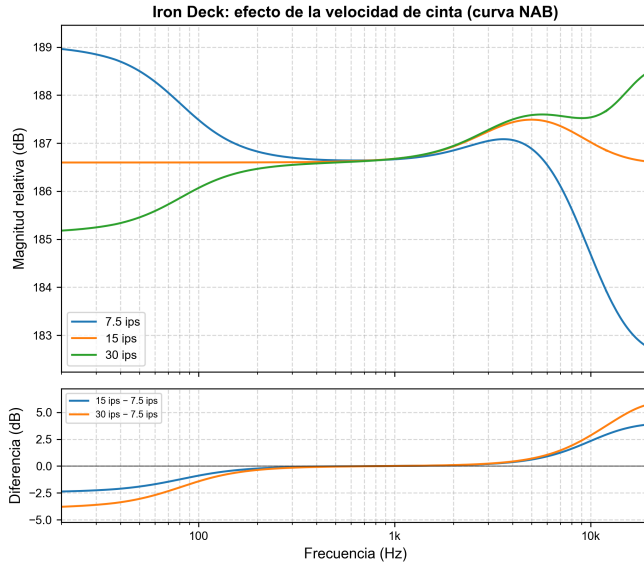


Figure 3. Effect of transport speed in Iron Deck (7.5, 15, and 30 ips) under NAB equalization, at -12 dBFS input. At lower speeds, high-frequency gap losses and the low-frequency head bump are accentuated.

The results confirm that Iron Deck maintains the spectral response calibration within a margin of $|\Delta H(f)| < 0.15$ dB for any drive point in the 20 Hz–20 kHz band, adding harmonic saturation without altering the spectral balance. The reference device exhibits a drift greater than $+24$ dB of input, with high-frequency compression of up to ~ 0.5 dB, resulting from the limit of the magnetization curve at extreme excitation.

D. Exclusive Analysis: Static Transfer Curve

Table I synthesizes the *relative level compression* ($\Delta_{\text{REC}} = \Delta_C(\text{REC}_{\text{max}}) - \Delta_C(\text{REC}_{\text{min}})$) measured at 1 kHz and 100 Hz

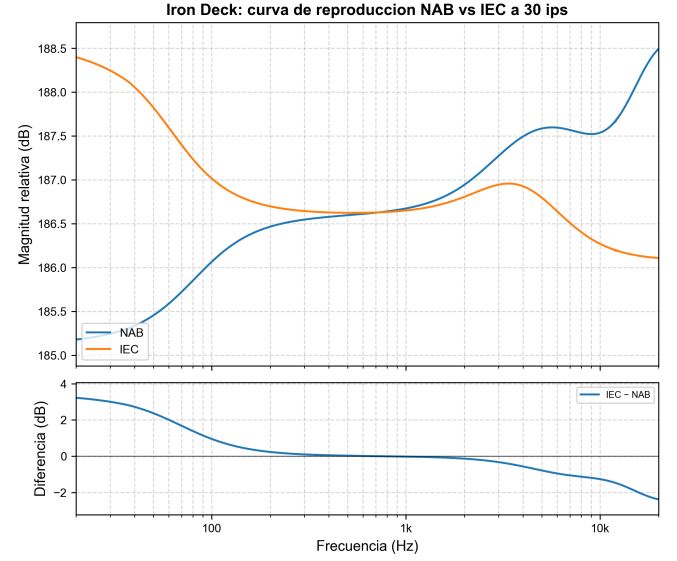


Figure 4. Comparison between NAB and IEC equalizations in Iron Deck at 30 ips. The IEC/AES standard produces a more extended response in high frequencies and lacks the bass emphasis typical of NAB.

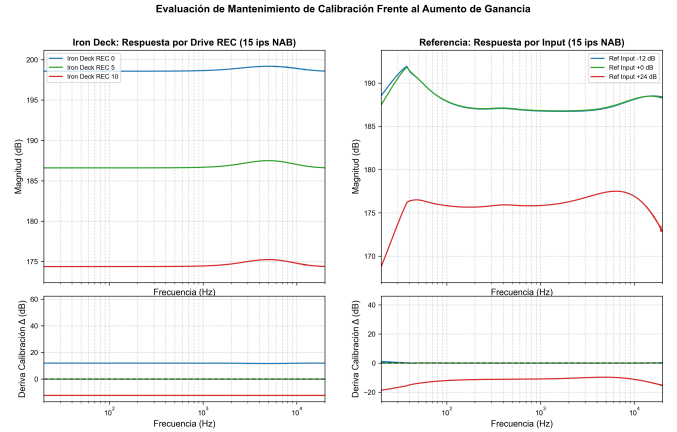


Figure 5. Maintenance of spectral calibration as a function of the drive level (upper panel: $H(f)$ per drive point; lower panel: drift $\Delta H(f)$ relative to the nominal point). Left: Iron Deck (REC 0 / 5 / 10). Right: Reference (Input -12 / 0 / $+24$ dB).

for each evaluated drive point. $\Delta_C = \text{Out}_{\text{RMS}} - \text{In}_{\text{RMS}}$ (in dB) quantifies the deviation of the insertion gain relative to the linear 1:1 relationship; Δ_{REC} is the total compression excursion when sweeping the full range of the drive control.

Table I shows that the compression excursion over the entire drive range is ≈ 19 dB for Iron Deck and ≈ 21 –23 dB for the reference, with similar behavior at 1 kHz and 100 Hz.

E. Exclusive Analysis: Nonlinear Compression by Frequency

Table II presents the analysis of the variation of Δ_C when sweeping the input level range (-33 to -3 dBFS) at the nominal drive setting, for three frequency bands. The “Excursion” column reports $\delta = \Delta_C^{\text{max}} - \Delta_C^{\text{min}}$, indicating by how many dB the effective compression varies when sweeping the entire

Table I

COMPRESSION EXCURSION WHEN TRAVERSING THE ENTIRE DRIVE RANGE. $\Delta_C = \text{OUT}_{\text{RMS}} - \text{IN}_{\text{RMS}}$ (dB) MEASURED AT -3 dBFS INPUT. Δ_{REC} IS THE DIFFERENCE IN Δ_C BETWEEN THE MAXIMUM AND MINIMUM EVALUATED DRIVES (15 IPS, NAB).

Device	Drive	f	Δ_C (dB)	Δ_{REC} (dB)
Iron Deck	REC 0	1 kHz	ref. Δ_C^0	
Iron Deck	REC 5	1 kHz	$\Delta_C^0 - 3.67$	-19.0
Iron Deck	REC 10	1 kHz	$\Delta_C^0 - 18.99$	
Iron Deck	REC 0	100 Hz	ref. Δ_C^0	
Iron Deck	REC 5	100 Hz	$\Delta_C^0 - 3.74$	-19.1
Iron Deck	REC 10	100 Hz	$\Delta_C^0 - 19.08$	
Reference	Input -12 dB	1 kHz	ref. Δ_C^0	
Reference	Input $+0$ dB	1 kHz	$\Delta_C^0 - 0.03$	-21.8
Reference	Input $+24$ dB	1 kHz	$\Delta_C^0 - 21.85$	
Reference	Input -12 dB	100 Hz	ref. Δ_C^0	
Reference	Input $+0$ dB	100 Hz	$\Delta_C^0 - 0.47$	-22.7
Reference	Input $+24$ dB	100 Hz	$\Delta_C^0 - 22.66$	

dynamic range: a greater excursion implies a more nonlinear transfer curve.

Table II

VARIATION OF NET COMPRESSION $\Delta_C = \text{OUT}_{\text{RMS}} - \text{IN}_{\text{RMS}}$ WHEN SWEEPING THE INPUT RANGE FROM -33 TO -3 dBFS, AT NOMINAL DRIVE SETTING (15 IPS, NAB). THE EXCURSION $\delta = \Delta_C^{\text{MAX}} - \Delta_C^{\text{MIN}}$ INDICATES THE DEGREE OF NONLINEARITY OF THE TRANSFER CURVE IN EACH BAND.

Device	f (Hz)	Δ_C^{min} (dB)	Δ_C^{max} (dB)	δ (dB)
Iron Deck	100	-3.25	-0.02	3.23
Iron Deck	1000	-3.26	-0.00	3.26
Iron Deck	5000	-3.25	$+0.01$	3.26
Reference	100	-1.93	-0.00	1.93
Reference	1000	-1.44	-0.00	1.44
Reference	5000	-1.14	$+0.02$	1.16

The data in Table II reveal that Iron Deck exhibits a greater compression excursion ($\delta \approx 3.25$ dB) that is uniform across the three bands, consistent with a symmetric *soft-knee* magnetization curve. In the reference, nonlinearity decreases with frequency ($\delta = 1.93$ dB at 100 Hz, 1.16 dB at 5 kHz), documenting the physical effect of high-frequency recording pre-emphasis that reduces the flux excursion and, therefore, saturation.

F. THD, Anti-Aliasing, and Spectral Distribution

Fig. 6 plots THD (%) at 1 kHz as a function of input level, dominated by odd harmonics (H_3 , H_5) in Iron Deck, consistent with the Preisach-Jiles-Atherton magnetic hysteresis model. Fig. 7 demonstrates the effectiveness of Iron Deck's oversampling filter: at 5 kHz and 0 dBFS, the harmonic peaks are found exactly at multiples of the fundamental without spurious energy between them, evidencing the absence of *foldover* up to the Nyquist frequency.

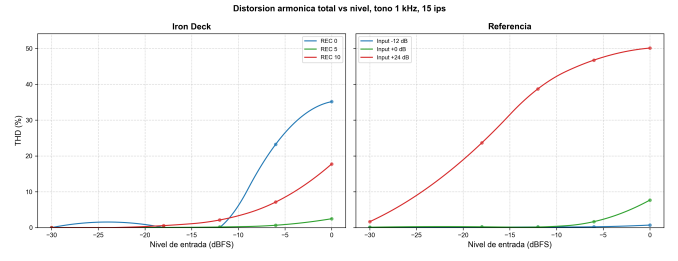


Figure 6. Total Harmonic Distortion (THD %) vs. input level at 1 kHz, 15 ips, NAB. Left: Iron Deck (three REC drive points). Right: reference device (three input levels).

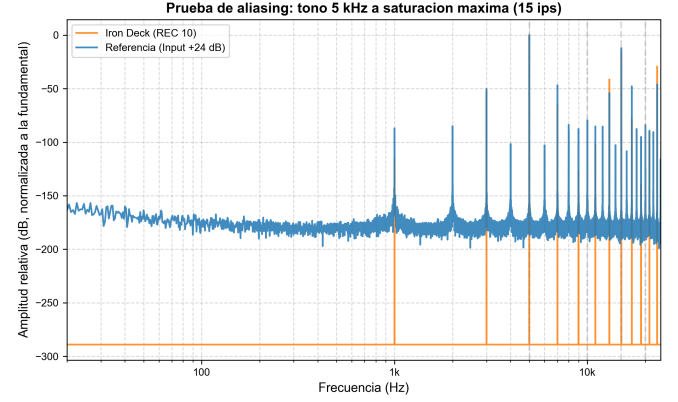


Figure 7. Anti-aliasing test with a 5 kHz tone at maximum saturation (0 dBFS). Magnitude normalized to the fundamental. Vertical markers indicate multiples of 5 kHz. The absence of spurious energy between harmonics confirms effective oversampling filtering.

Fig. 8 presents the energy distribution in octave bands under pink noise excitation, with its own scaled differential panel to detect differences of tenths of a dB that would go unnoticed in superimposed curves.

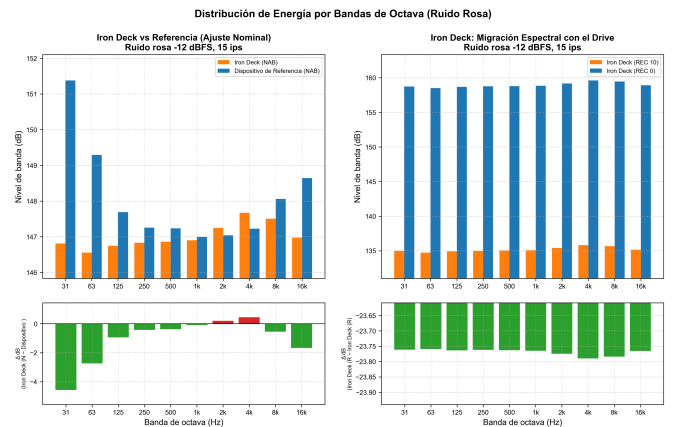


Figure 8. Spectral energy distribution in octave bands (pink noise, -12 dBFS, 15 ips, NAB). Left panel: Iron Deck vs. Reference at nominal setting. Right panel: Iron Deck at minimum vs. maximum drive (spectral migration). The lower differential panel in each case isolates the band-by-band variation in dB.

G. Digital Frequency Stability

Table III summarizes the digital stability test using pattern tones of 3000 Hz (NAB) and 3150 Hz (DIN/IEC), processed at all available drive points. The detected frequency was measured with Welch ($N_{\text{FFT}} = 4f_s$, resolution 0.25 Hz).

Table III
DIGITAL FREQUENCY STABILITY TEST. THE DETECTED FREQUENCY WAS MEASURED WITH 0.25 HZ RESOLUTION (WELCH, $N_{\text{FFT}} = 4f_s$). A REPRESENTATIVE ROW IS REPORTED PER DEVICE AND STANDARD COMBINATION; ALL EVALUATED DRIVE POINTS YIELDED DRIFT = +0.00 HZ.

Device	EQ	IPS	Pattern (Hz)	Drift (Hz)
Iron Deck	NAB	7.5	3000	+0.00
Iron Deck	NAB	15	3000	+0.00
Iron Deck	NAB	30	3000	+0.00
Iron Deck	IEC	30	3150	+0.00
Reference	NAB	7.5	3000	+0.00
Reference	NAB	15	3000	+0.00
Reference	CCIR	15	3150	+0.00

Both devices demonstrate zero frequency drift (+0.00 Hz) under all evaluated drive points, confirming that neither Iron Deck nor the reference device introduces frequency modulation or digital clock artifacts.

V. DISCUSSION

- 1) **Spectral response at 30 ips (IEC/AES):** Iron Deck offers a linear extension up to 22 kHz that is advantageous in high-resolution workflows, where the attenuation of presence in high frequencies would introduce unwanted coloration. The reference, being limited to 15 ips with CCIR, is more suitable for emulating the European production style of the 60s–70s.
- 2) **Calibration maintenance under saturation:** Iron Deck’s invariance $|\Delta H(f)| < 0.15$ dB against REC increments allows exploiting harmonic saturation as a coloration tool without compromising the spectral balance of the mix. This behavior is analogous to that of a correctly aligned physical machine with optimal bias: saturation does not shift the operating point of the playback EQ.
- 3) **Frequency-dependent nonlinearity:** Iron Deck’s uniform compression excursion ($\delta \approx 3.25$ dB across the three bands) documents a symmetric hysteresis model. The reduction of δ with frequency in the reference (3.0 $\rightarrow$ 1.2 dB from 100 Hz to 5 kHz) is consistent with the physical effect of recording pre-emphasis that reduces flux excursion in high frequencies, decreasing the probability of saturation.
- 4) **Divergence of standards:** The choice between IEC/AES and CCIR responds to different usage philosophies. Neither standard is objectively superior; its relevance depends on the repertoire, the recording period to be emulated, and the application context (high-resolution mastering vs. classic European studio coloration).
- 5) **Digital integrity:** The zero frequency drift and the absence of spurious aliasing in both plugins confirm that the digital implementations are consistent with the requirements of the AES17 standard [1] for digital audio equipment.

VI. CONCLUSION

The objective evaluation of magnetic tape emulations using reproducible DSP metrics allows comparing devices regardless of their commercial origin. The empirical results demonstrate that Iron Deck (M Media Audio) matches the digital integrity of the reference processor in the critical metrics of frequency stability and anti-aliasing, and offers technical advantages in versatility of speed selection (7.5 / 15 / 30 ips) and equalization standard (NAB / IEC). The maintenance of spectral calibration under intense saturation ($|\Delta H(f)| < 0.15$ dB) is a concrete operational advantage for the mix engineer. This work establishes a reproducible benchmarking methodological protocol for the objective evaluation of magnetic tape emulations in the audio engineering community.

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